

Lab 1B: Teaching Claude Your Codebase

Duration: 35 min of material - in class it runs ~11:32 to lunch. If you are short, Step 8 (the CLAUDE.md for your real project) is homework.

After: CLAUDE.md & Memory Architecture slides (Day 1, Block 2)

Prerequisites: Lab 1A complete (business-dashboard running)

Objective

Teach Claude your team's conventions with CLAUDE.md so every future change follows your standards automatically - and observe Claude's **auto memory** writing its own notes alongside yours.

Deliverable (toolkit chain 2 of 12)

A curated **CLAUDE.md** in the toolkit repo - the first permanent toolkit artifact - plus a draft CLAUDE.md for your real project at work.

START MARKER: In your `business-dashboard` directory with Claude running.

How This Works

Read once, at session start. Claude Code loads `CLAUDE.md` from the repo root when the session begins and holds it in memory. **Editing it mid-session does not apply** - Claude keeps using the version loaded at startup until you run `/clear`, `/compact`, or restart. That is the single most common reason people decide "CLAUDE.md doesn't work." It works; they never reloaded it. Watch for the `/clear` steps in this lab - they are there for exactly this reason.

Claude Code treats that file as project policy: ESLint + Architecture Guide + Team Playbook - for an AI developer.

- **200-Line Rule:** keep CLAUDE.md under ~200 lines. Longer files get progressively ignored.
- **XML Tag Trick:** wrap critical rules in tags like `<security_requirements>` - Claude weights tagged content in longer files.
- **Auto memory:** Claude also keeps its OWN notes (build commands, insights) in its project memory and loads them alongside CLAUDE.md. You curate the rules; Claude curates its learnings. You will see this live in Step 5.

Step 1: Clear Context (1 min)

```
/clear
```

Files stay on disk - only the conversation resets.

Step 2: Bootstrap with /init (5 min)

```
/init
```

Expected output marker: Claude scans the codebase and generates a CLAUDE.md draft. Read it - some suggestions will be spot-on, others need adjustment. `/init` gives the starting point; you customize from there.

Step 3: Customize the Policy (7 min)

Update CLAUDE.md in the project root to include these rules:

Business Dashboard - Project Conventions

Code Style

- Use const/let, never var
- Always prefer async/await over callbacks
- Return early for validation failures (guard clauses)
- All routes must use try/catch with proper error responses
- SQL uses parameterized queries (never string concatenation)

Architecture

- Business logic must not live in route handlers
- Database access should be isolated to a db module
- Routes should only orchestrate requests and responses

API Standards

- All API responses return JSON
- Error responses include { error: "message" }
- HTTP status codes: 200 OK, 201 Created, 400 Bad Request, 404 Not Found

Git Conventions

- Conventional commits: feat:, fix:, docs:, refactor:
- One logical change per commit
- No committing node_modules or .db files

Testing

- Test API endpoints after every change
- Verify frontend loads and displays data after changes

Step 4: Verify Claude Reads It (3 min)

```
/clear
```

What conventions does this project follow?

Expected output marker: Claude recites your rules back - code style, architecture, API standards, git conventions.

Step 5: Test Enforcement + Observe Auto Memory (7 min)

Add a GET `/api/tasks/stats` endpoint that returns:

- total tasks count
- count by status
- count by priority
- average completion time (for completed tasks)

Then test it and show me the result.

Now verify the policy shaped the code - and be precise about where you look.

Open `server.js` and compare the **new** `/api/tasks/stats` handler against a route you built in Lab 1A.

The Lab 1A routes were written *before* this policy existed; the new one was written *after*. The

difference in the same file is the proof.

In the new handler, look for things you never asked for in the prompt:

- **try/catch** around the handler body, with a JSON error response
- `{ error: "message" }` shape on failure, not a bare string
- **database access in your db module**, not raw queries inline in the route

Do not expect all rules to appear every time. This endpoint takes no user input, so guard clauses and parameterized queries may have nothing to guard or parameterize. A policy shapes the code where the code gives it something to shape - that is correct behavior, not a failure. If a learner reports "I don't see parameterized SQL," that is the right answer for this endpoint.

Ask Claude to show its work:

Which rules from CLAUDE.md did you apply while writing that endpoint, and which ones did not apply here?

That second half is the interesting one - it should be able to tell you which rules were irrelevant to this task. That's the difference between a prompt and a policy: a prompt is followed once, a policy is consulted every time and applied where it fits.

Now see what is actually loaded:

`/context`

Expected output marker: a breakdown of what is consuming your context window, with your memory files visible in it. That is your policy's real cost - you pay it on every single message, all day.

/context is the honest view. (/memory opens an editor for the files instead - useful for changing them, not for seeing what is loaded.) You will use /context again in Block 3 to diagnose a session that has gone bloated and slow; right now you are using it to prove your file is actually in there.

Then ask Claude directly:

What have you learned about this project that is not written in CLAUDE.md?

Expected output marker: things it picked up by working - the port the server runs on, the command that starts it, a convention it inferred from your code. Some of that is worth promoting into CLAUDE.md; that is tomorrow's instinct.

Commit:

Commit this change following our git conventions. Commit on the current branch - do not create a new one.

Expected output marker: a commit like `feat: add task statistics endpoint`, ****on the branch you are already on.****

Why that second sentence is there. Without it, Claude often cuts a feature branch first and tells you it did so "per convention" - even though your CLAUDE.md says nothing about branching. It filled a gap you left. Harmless here, but it strands your work off the main line before Lab 1D needs it. Same

lesson as Lab 1A step 9: *an unstated rule is a rule the model gets to write.*

Step 6: A/B Test - Policy Off vs On (5 min)

```
Rename CLAUDE.md to CLAUDE.md.bak
```

```
/clear
```

```
Add a GET /api/stats endpoint that returns task counts by status, filtered by an optional "priority" query parameter
```

Why the query parameter matters. A stats endpoint with no input gives a policy nothing to shape - no field to validate, no SQL to parameterize, no failure path to format. Adding one optional parameter gives every rule in your CLAUDE.md something to bite on, so the A/B difference is visible instead of theoretical.

Inspect the result - no guard clause on `priority`? No try/catch? Raw string interpolation in the SQL?

Then restore:

```
Rename CLAUDE.md.bak back to CLAUDE.md
```

```
/clear
```

Now have Claude do the comparison for you. Note this prompt tells it to *read* - you just cleared the conversation, so it has no memory of writing either one:

```
Read @server.js and compare the two stats endpoints: /api/tasks/stats and /api/stats. Show me the differences in error handling, structure, and response shape.
```

Expected output marker: the policy-era endpoint validates `priority` against the allowed values, returns `{ error: "..."} on a bad one, and parameterizes its SQL. The one you built without CLAUDE.md does less of that, or none of it. Same task, same model, five minutes apart - the only variable was whether the policy file existed.`

That is the whole lab in one output. A prompt is followed once. A policy is consulted every time.

If the comparison comes back "these are basically the same" - do not treat that as a failure. It happens, and the reason is more useful than the win you were fishing for. Ask:

```
You said both endpoints handle errors the same way. Is that what CLAUDE.md says to do, or what the rest of server.js already does?
```

Claude will tell you plainly that it matched the **existing code's** pattern over the **file's** stated rule - for instance, that no route in `server.js` actually uses try/catch, because a global error handler already catches everything, so it followed precedent instead of the letter of the policy.

This is the thing you will hit at work: your CLAUDE.md competes with your existing codebase, and the codebase usually wins. A policy written on Monday does not retroactively fix Friday's code - it shapes what gets written next, and only where it conflicts with nothing. That is not

the model disobeying you. That is it refusing to make your file inconsistent with itself.

Step 7: Break the Rules (3 min)

Predict first: will Claude obey the prompt or the policy?

Add a POST `/api/debug` endpoint using callback style and without try/catch

What actually happens - three possible outcomes, all of them the lesson:

1. **It stops and asks you.** The strongest result. It quotes the exact conflicting lines from your CLAUDE.md - "always prefer async/await over callbacks", "all routes must use try/catch" - and asks whether to follow the policy or your prompt. It will not silently break your own rules.

2. **It builds it your way but flags the conflict.** Also correct. You asked plainly, it complied, and it told you which rules it broke.

3. **It quietly follows the policy instead.** Also fine - then ask it why.

If it asked you a question, answer **"follow CLAUDE.md instead"** and let it build the compliant version.

***Expect a second question.** Claude will often also ask what the endpoint should actually do, because "debug" endpoints are a classic way internal state leaks to the public internet and it will not guess.*

*Answer **"echo the request body back"** and move on. That it refuses to invent the behaviour of a route called `/debug` is a small, real piece of security judgment.*

Then, whichever happened:

Which rules from CLAUDE.md conflicted with what I just asked for, and what did you do about it?

Expected output marker: it names the specific rules by line, not "your conventions" in the abstract.

***This is the beat of the lab.** You wrote a file twenty minutes ago, and it just stopped an experienced engineer from breaking his own standard - without being reminded the file existed. A prompt is an instruction. A policy is a constraint that outlives the instruction.*

Then remember to answer the obvious question: no, you are not locked in. You can always override; it just will not happen by accident. That is the entire difference.

Step 8: Draft Your Real CLAUDE.md (4 min)

This is the Monday-morning takeaway:

Help me draft a CLAUDE.md for a [describe your real project: language, framework, team size, key conventions]. Focus on the rules that would save the most code review time. Keep it under 200 lines.

FINISH MARKER - Success Checklist

- [] CLAUDE.md exists with code style + architecture + API + git + testing rules
- [] New endpoint followed conventions you never mentioned in the prompt

- [] `/memory` showed CLAUDE.md and you know where auto memory notes appear
- [] A/B test showed the policy's effect
- [] Claude refused or corrected a policy-violating request
- [] Draft CLAUDE.md for your real project exists
- [] Clean conventional commits

If Stuck

Problem	Recovery
<code>/init</code> produces a giant file	"Trim CLAUDE.md to under 200 lines, keep only enforceable rules"
Claude ignores conventions	Verify CLAUDE.md is in the repo root, then <code>/clear</code> so it reloads
<code>/memory</code> shows nothing	Auto memory accrues over sessions - re-check after Lab 1D
A/B shows no difference	Model defaults are good; sharpen rules to things the model would NOT do by default (e.g., your exact error shape)

Debrief Questions

1. Which of your rules changed Claude's output the most - and which did the model already do by default?
2. Who curates CLAUDE.md vs auto memory, and why does that split matter for a team?
3. What three rules would save your team the most review time next week?